



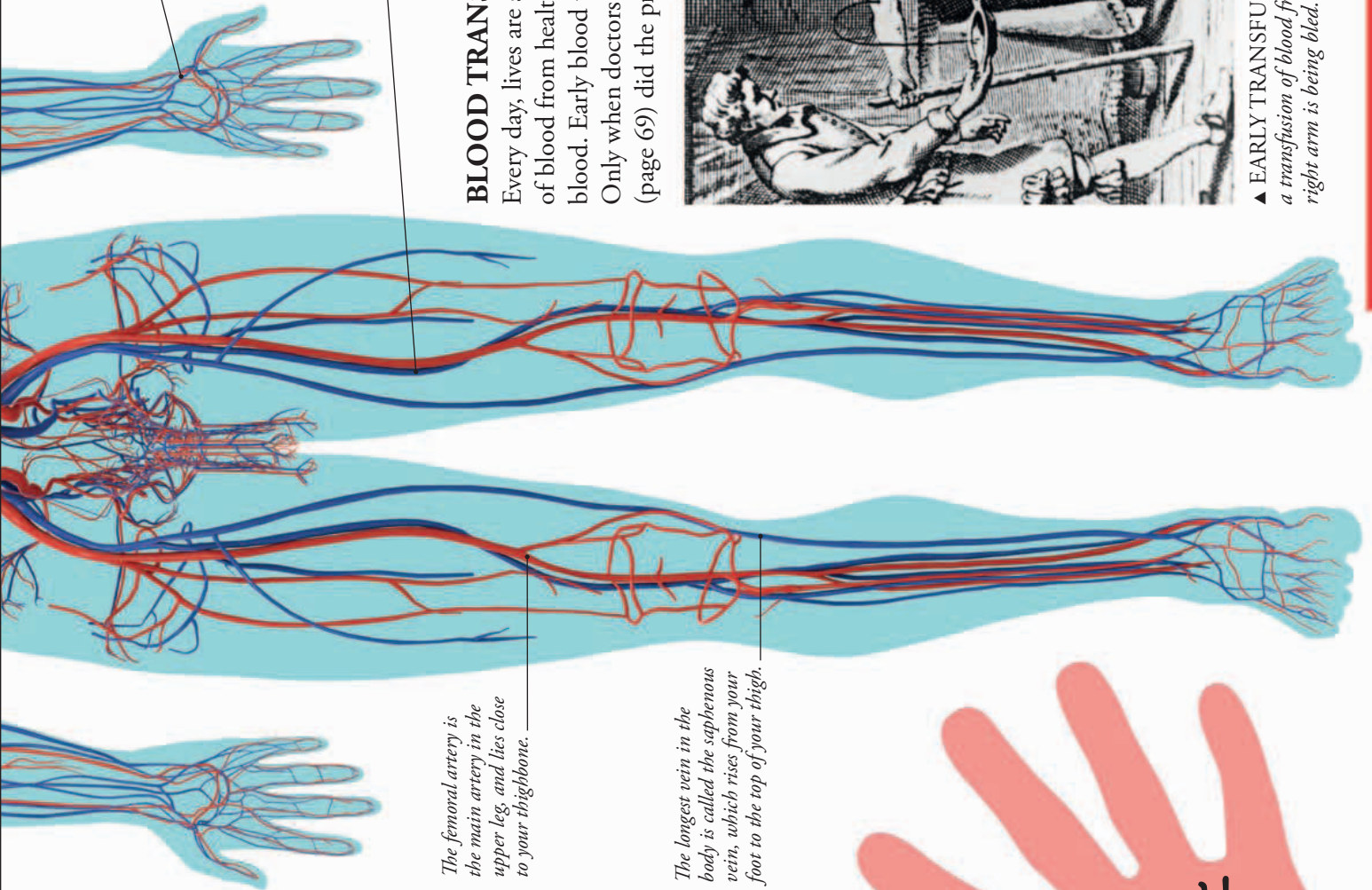
Since ancient times, no one knew how the heart and blood worked. The riddle was finally solved in the 1600s by English doctor William Harvey. He carried out various experiments, including tying a tight band around a person's upper arm and studying blood flow through the veins (shown above). Harvey discovered that blood circulated around the body, pumped by the heart.

BLOOD TRANSFUSIONS

Every day, lives are saved by the transfer, or transfusion, of blood from healthy donors to patients who have lost blood. Early blood transfusions often resulted in death. Only when doctors became aware of blood types (page 69) did the process become safer.



▲ **EARLY TRANSFUSION** This 17th-century picture shows a transfusion of blood from a dog (far right) to a man, whose right arm is being bled.



Blood vessels stretch into every corner of your hand, forming a loop for the return journey to the heart.

The femoral vein drains blood from your thigh, carrying it your heart.

The femoral artery is the main artery in the upper leg, and lies close to your thighbone.

The longest vein in the body is called the saphenous vein, which rises from your foot to the top of your thigh.

